1. Creating Hello World Application

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

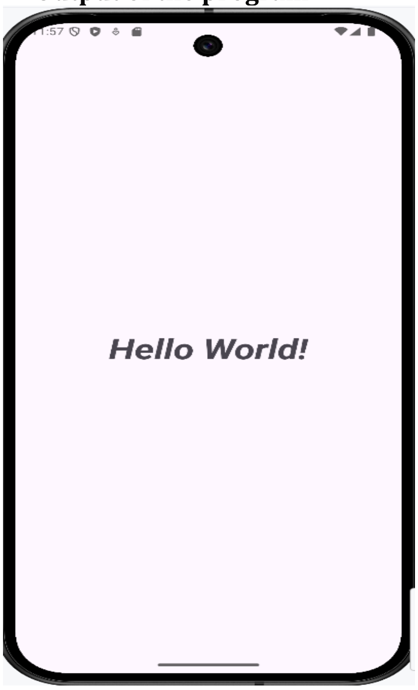
=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World!"
      android:textSize="40sp"
      android:textStyle="bold|italic"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

=>Coding part of MainActivity.java

```
package com.example.loginapplication;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
   }
}
```

=>**Output of the program**



2. Creating an Application that displays message based on screen orientation

Step 1: Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="This is Portrait"
      android:textSize="40sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.496"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.25" />

   <Button
      android:id="@+id/button"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:onClick="onClick"
      android:text="Launch new activity"
      app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.499" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

=>Coding part of MainActivity.java

```
package com.example.orientation;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
    }

    public void onClick(View v){
        Intent intent=new Intent(MainActivity.this,Nextactivity.class);
        startActivity(intent);
    }

}
```

Step 2: Create another new empty views activity and give the name as Nextactivity (Go to app>> New>>Activity>>Empty Views Activity)

=>Coding part of Activity_nextactivity.xml

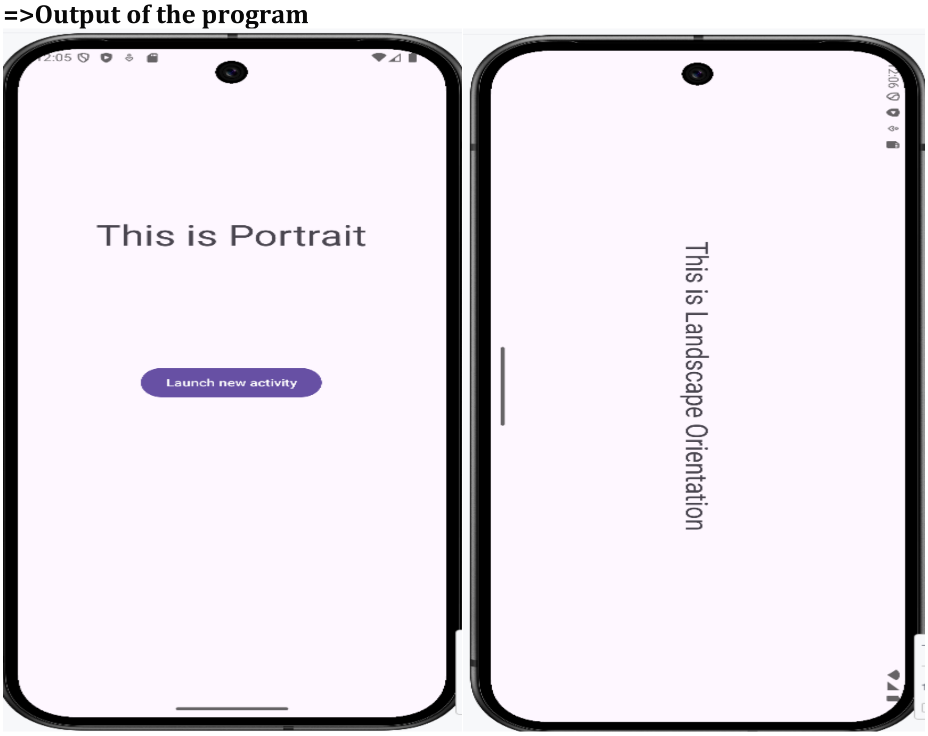
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Nextactivity">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is Landscape Orientation"
        android:textSize="30sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 3: Add the screen orientation values in AndroidManifest.xml

=>Coding part of AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Orientation"
        tools:targetApi="31">
        <activity
            android:name=".Nextactivity"
            android:exported="false"
            android:screenOrientation="landscape"

            />
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:screenOrientation="portrait"
            >
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

=>Output of the program



3. Create and Application to develop Login window using UI controls

Step 1: Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

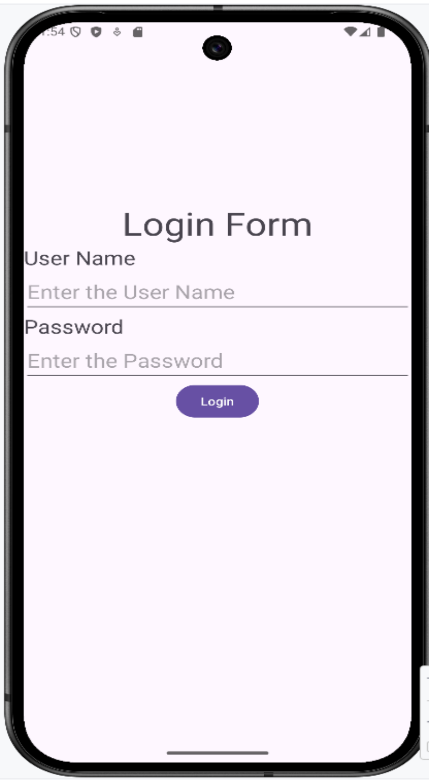
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">

  <TextView
    android:id="@+id/tvTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:layout_marginTop="200dp"
    android:text="Login Form"
    android:textSize="40sp" />

  <TextView
    android:id="@+id/tvUserName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="User Name"
    android:textSize="24sp" />

  <EditText
    android:id="@+id/etUserName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the User Name"
    android:inputType="text"
    android:textSize="24sp" />

  <TextView
    android:id="@+id/tvPassword"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Password"
    android:textSize="24sp" />

  <EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the Password"
    android:inputType="textPassword"
    android:textSize="24sp" />

  <Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
```

```
    android:layout_gravity="center" />

</LinearLayout>
```

=>Coding part of MainActivity.java

```java
package com.example.loginapplication;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
  }
}
```

=>Output of the program



4. Create and Application to implement new activity using explicit intent and implicit intent

Step 1: Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Explicit Activity"
        android:textSize="30sp"
        android:layout_gravity="center"
        android:layout_marginTop="200dp"
        />

    <Button
        android:id="@+id/btnExplicitContent"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="200dp"
        android:onClick="onClick"
        android:text="Explicit Activity" />

</LinearLayout>
```

=>Coding part of MainActivity.java

```java
package com.example.implicitexplicit;
import android.content.Intent;
import android.view.View;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;



public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
```

```java
    public void onClick(View view) {
        Intent intent = new Intent(MainActivity.this, NextActivity.class);
        startActivity(intent);
    }

}
```

Step 2: Create another new empty views activity and give the name as Nextactivity (Go to app>> New>>Activity>>Empty Views Activity)

Coding part of activity_next.xml

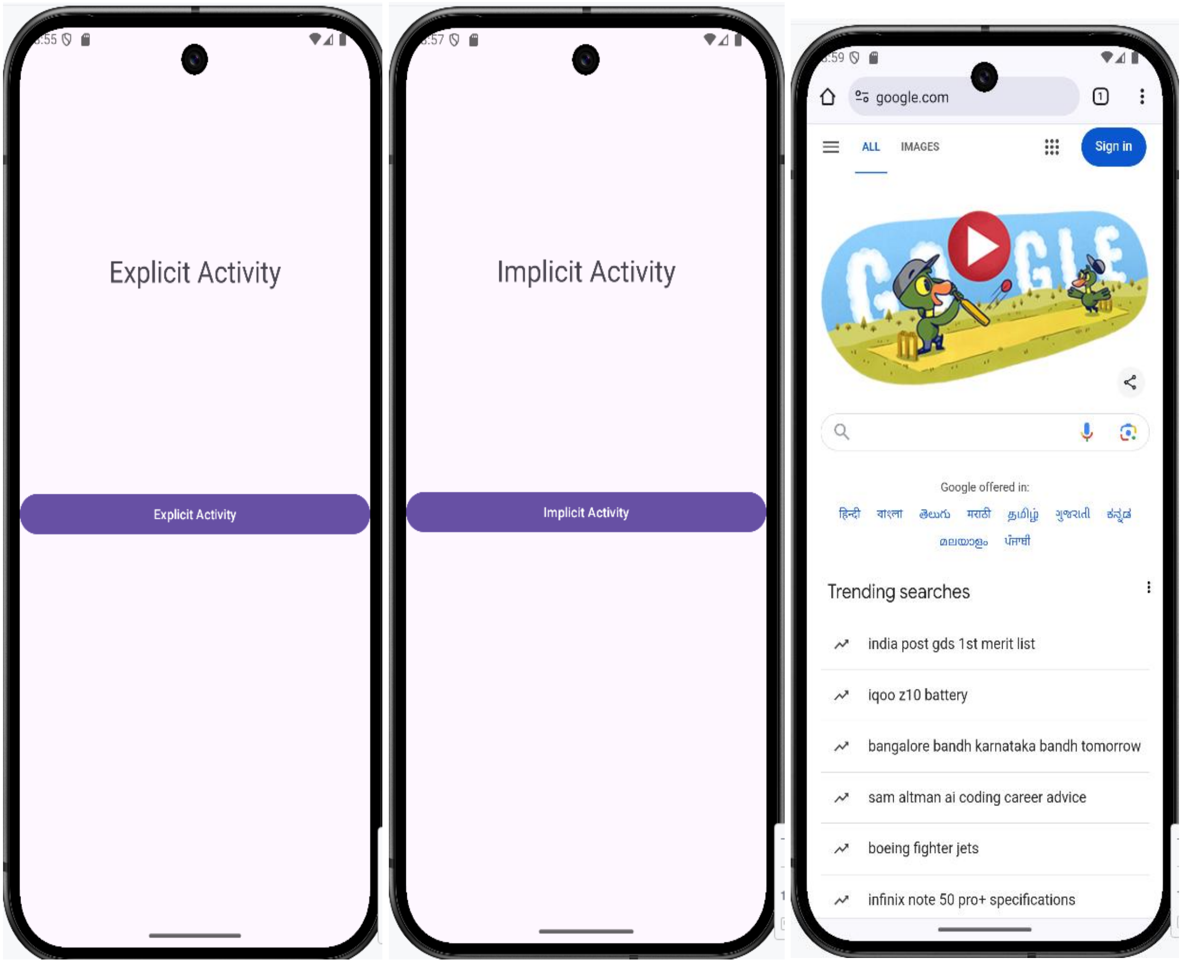
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".NextActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Implicit Activity"
        android:textSize="30sp"
        android:layout_gravity="center"
        android:layout_marginTop="200dp"
        />

    <Button
        android:id="@+id/btnImplicitContent"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="200dp"
        android:onClick="onClick"
        android:text="Implicit Activity" />

</LinearLayout>
```

=>Coding part of NextActivity.java

```java
package com.example.implicitexplicit;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.net.Uri;
import android.view.View;


public class NextActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_next);
```

```
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

    }


    public void onClick(View view) {
        Uri webpage= Uri.parse("http://www.google.com");
        Intent intent = new Intent(Intent.ACTION_VIEW, webpage);
        startActivity(intent);
    }


}
```

=>Output of the program



// 5 Create an application that displays custom designed Opening Screen

Step1. Project creation Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.

Next follow the below steps to create background image files

Step 2.Add 2 images to the project

a. Copy the image files (Ctrl +C) from the file location

b. Click app>>res>> drawable>> right click Paste(Ctrl+V)

Step 3. Change the activity_main.xml file and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout  xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:background="@drawable/android1" >
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome !!!"
        android:textSize="40sp"
        android:textStyle="bold"
        android:layout_centerInParent="true" />
</RelativeLayout>
```

=>Coding part of MainActivity.java

```
package com.example.customdesign;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.os.Handler;


public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        Handler handler=new Handler();
        handler.postDelayed(new Runnable() {
            @Override
            public void run() {
```

```
            Intent intent=new Intent(MainActivity.this,NextActivity.class);
            startActivity(intent);
            finish();
            }
        },3000);

    }
}
```

Step 4. Create one more empty views activity and give the name as NextActivity(Go to app>> New>>Activity>>Empty Views Activity) and change the activity_next.xml file and *no changes* in the NextActivity.java file

Coding part of activity_next.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout  xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:app2="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".NextActivity"
    android:background="@drawable/android3" >
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to home Page"
        android:textColor="@color/black"
        android:textSize="32sp"
        android:textStyle="bold"
        android:layout_centerInParent="true" />
</RelativeLayout>
```

=>Coding part of NextActivity.java

```
package com.example.customdesign;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class NextActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_next);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

=>Output of the program

6. Create and Application to show all views

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:fillViewport ="true"
  tools:context=".MainActivity">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">

  <TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="This is a TextView"
    android:textSize="18sp"
    android:layout_marginBottom="16dp"/>

  <EditText
    android:id="@+id/editText"
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:hint="Enter your text here"
    android:layout_marginBottom="16dp" />

  <Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Me"
    android:layout_marginBottom="16dp" />

  <CheckBox
    android:id="@+id/checkbox"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Check Me"
    android:layout_marginBottom="16dp" />
   <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Which Section You belong to"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
  <RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginBottom="16dp" />

  <RadioButton
    android:id="@+id/radioButton1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="A section" />

  <RadioButton
    android:id="@+id/radioButton2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="B Section" />

  <Switch
    android:id="@+id/switch1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Switch Testing"
    android:layout_marginBottom="16dp"/>
  <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select your favourite Programming Language"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
  <Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="16dp"/>
  <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

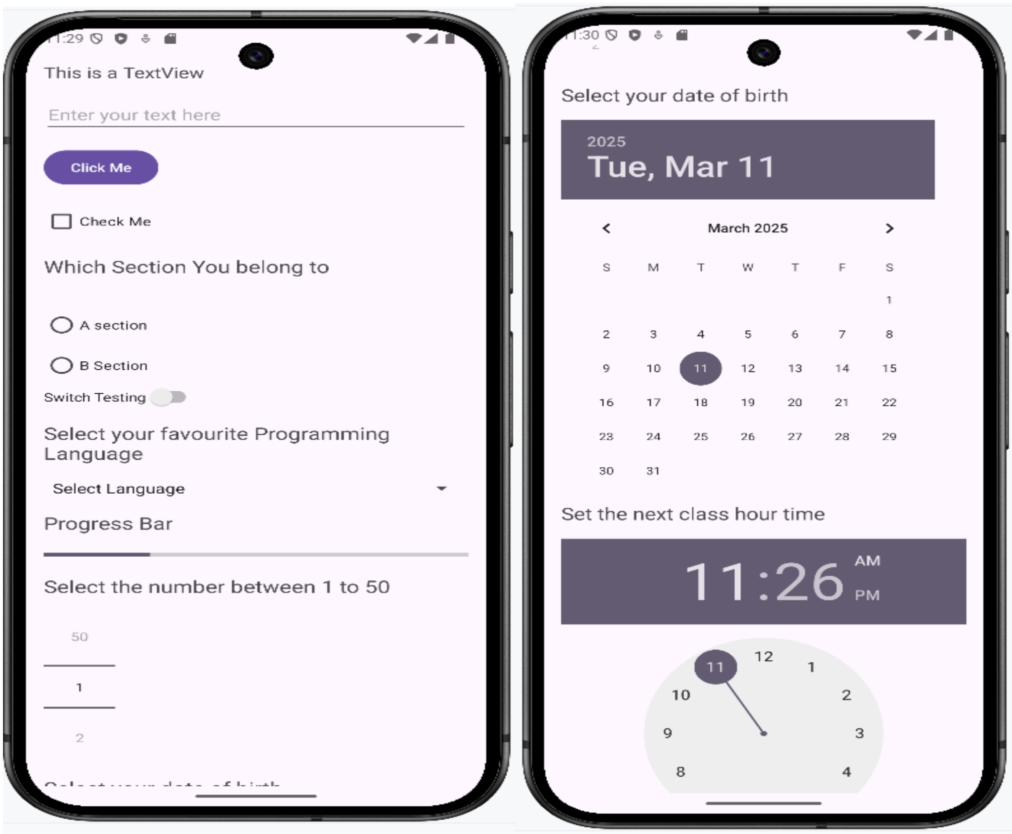
```xml
        android:text="Progress Bar"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
    <ProgressBar
        android:id="@+id/progressBar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        style="@style/Widget.AppCompat.ProgressBar.Horizontal"
        android:progress="25"
        android:layout_marginBottom="16dp" />
    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select the number between 1 to 50"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
     <NumberPicker
        android:id="@+id/numberPicker1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"/>
     <TextView
        android:id="@+id/textView5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select your date of birth"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
     <DatePicker
      android:id="@+id/datePicker"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginBottom="16dp"/>
    <TextView
        android:id="@+id/textView6"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set the next class hour time"
        android:textSize="20sp"
        android:layout_marginBottom="16dp"/>
    <TimePicker
      android:id="@+id/timePicker"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginBottom="16dp"/>

    </LinearLayout>

</ScrollView>
```

=>Coding part of MainActivity.java

```java
package com.example.mainactivity;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Spinner;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
import java.util.ArrayList;
import android.widget.NumberPicker;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        Spinner spinner = findViewById(R.id.spinner);
        ArrayList<String> arrayList = new ArrayList<>();
        arrayList.add("Select Language");
        arrayList.add("JAVA");
        arrayList.add("ANDROID");
        arrayList.add("C Language");
        arrayList.add("CPP Language");
        arrayList.add("Python Programming");

        ArrayAdapter<String> arrayAdapter = new ArrayAdapter<String>(this,
                android.R.layout.simple_spinner_item, arrayList);
         arrayAdapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        spinner.setAdapter(arrayAdapter);


        NumberPicker numberPicker1 = findViewById(R.id.numberPicker1);
        numberPicker1.setMinValue(1);
        numberPicker1.setMaxValue(50);

    }
}
```

=> Output of the program



7 Create a menu in application

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.

Steps to be followed to create the menu

Step1. Create Android resource directory by clicking res>>new>>Android Resource directory, give the name as menu

Step2. Right click menu folder click new>> Menu Resource File, give the name of the file as menus.

Add the following code in menus.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item android:id="@+id/php"  android:title="PHP"/>
   <item android:id="@+id/java"  android:title="JAVA"/>
   <item android:id="@+id/csharp"  android:title="C#"/>
</menu>
```

Step 3. =>Coding part of Activity_main.xml (Remove Helloworld textview control)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 4. =>Coding part of MainActivity.java

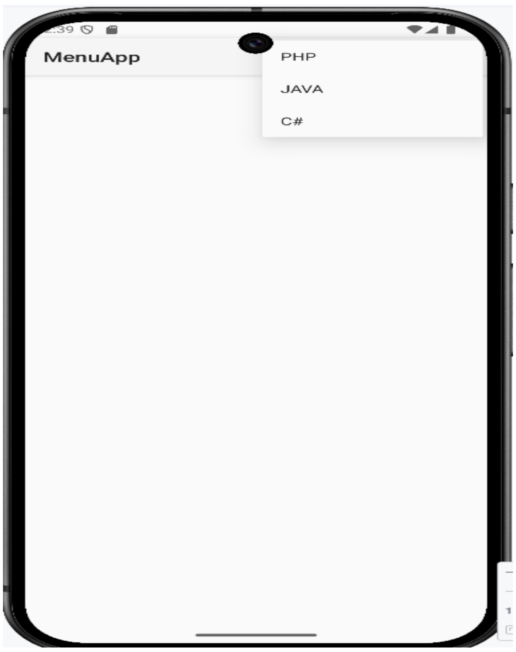
```
package com.example.menuapp;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;


public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
   }
```

```
@Override
public boolean onCreatePanelMenu(int featureId,Menu menu)
{
    MenuInflater inflater=getMenuInflater();
    inflater.inflate(R.menu.menus,menu);
    return true;
}
@Override
public boolean onOptionsItemSelected(MenuItem item)
{
    Toast.makeText(this, "You selected " + item.getTitle(), Toast.LENGTH_SHORT).show();
    return true;
}
```

Step 5. Change the theme value AndroidManifest.xml file

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

=>Output of the program



8. Create an application to read/ write the Local data.

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java
=>Coding part of Activity_main.xml

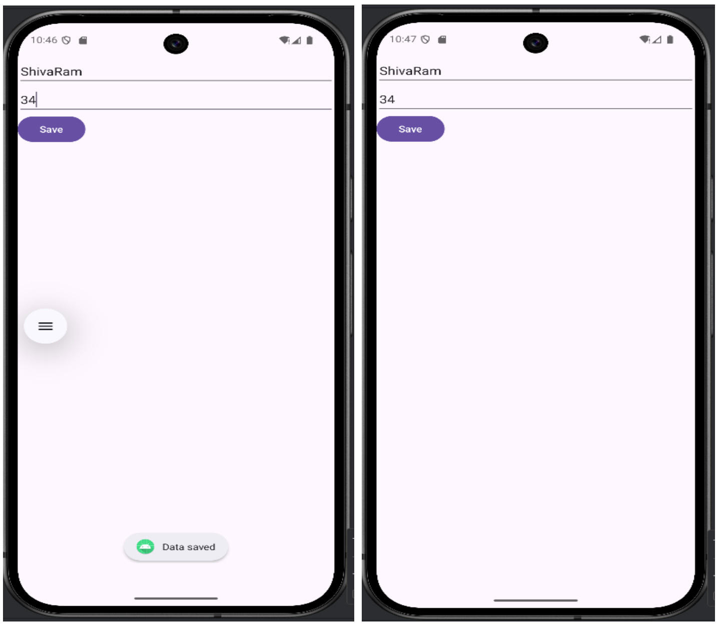
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">

  <EditText
    android:id="@+id/txtname"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="text"
    android:hint="Enter the Name"
  />

  <EditText
    android:id="@+id/txtage"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="text"
    android:hint="Enter the Age"
  />

  <Button
    android:id="@+id/btnsave"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="onClick"
    android:text="Save" />
</LinearLayout>
```

=>Coding part of MainActivity.java

```
package com.example.program8;

import android.content.Context;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    private EditText txtname,txtage;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

       txtname=findViewById(R.id.txtname);
       txtage=findViewById(R.id.txtage);
    }

  protected void onResume()
  {
      super.onResume();
      SharedPreferences sharedPreferences = getSharedPreferences("MyPrefs",
              Context.MODE_PRIVATE);
      String name = sharedPreferences.getString("name", "");
      String age = sharedPreferences.getString("age", "");
      txtname.setText(name);
      txtage.setText(age);
  }
  public void onClick(View v)
  {
      SharedPreferences sharedPreferences = getSharedPreferences("MyPrefs",
              Context.MODE_PRIVATE);
      SharedPreferences.Editor editor = sharedPreferences.edit();
      editor.putString("name",txtname.getText().toString());
      editor.putString("age", txtage.getText().toString());
      editor.apply();
      Toast.makeText(this, "Data saved", Toast.LENGTH_SHORT).show();
  }
}
```

=>Output of the program



//9 Create / Read / Write data with database (SQLite)

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent">
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="50dp"
    android:layout_y="20dp"
    android:text="Student Details"
    android:textSize="30sp" />

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="20dp"
    android:layout_y="110dp"
    android:text="Enter Rollno:"
    android:textSize="20sp" />

  <EditText
    android:id="@+id/Rollno"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="100dp"
    android:inputType="number"
    android:textSize="20sp" />
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="20dp"
    android:layout_y="160dp"
    android:text="Enter Name:"
    android:textSize="20sp" />

<EditText
    android:id="@+id/Name"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="150dp"
    android:inputType="text"
    android:textSize="20sp" />

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="20dp"
    android:layout_y="210dp"
    android:text="Enter Marks:"
    android:textSize="20sp" />

<EditText
    android:id="@+id/Marks"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="200dp"
    android:inputType="number"
    android:textSize="20sp" />

<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:onClick="onClickInsert"
    android:text="Insert"
    android:textSize="30dp" />

<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"
    android:onClick="onClickDelete"
    android:text="Delete"
    android:textSize="30dp" />

<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:onClick="onClickUpdate"
    android:text="Update"
    android:textSize="30dp" />

<Button
```

```
        android:id="@+id/View"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="200dp"
        android:layout_y="400dp"
        android:onClick="onClickView"
        android:text="View"
        android:textSize="30dp" />

    <Button
        android:id="@+id/ViewAll"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:layout_x="100dp"
        android:layout_y="500dp"
        android:onClick="onClickViewAll"
        android:text="View All"
        android:textSize="30dp" />

</AbsoluteLayout>
```

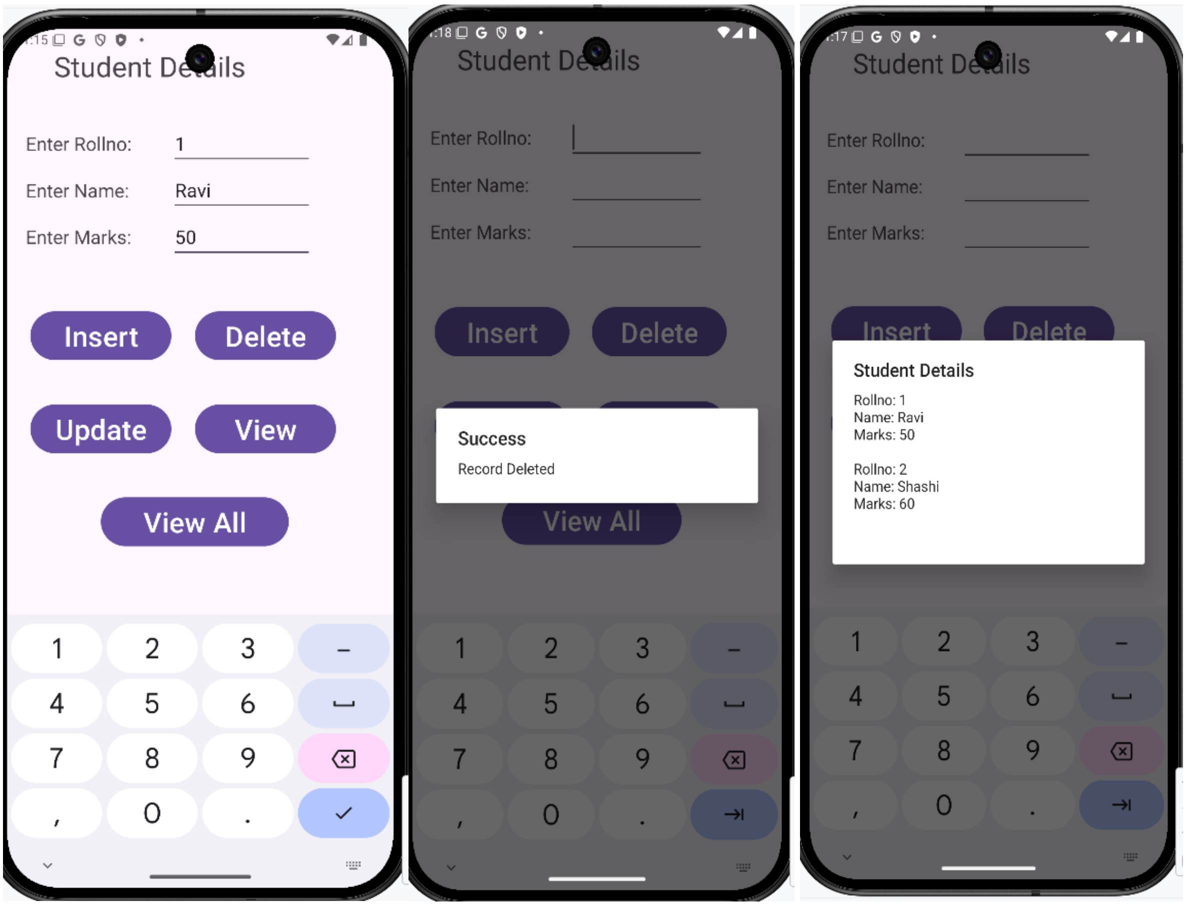
=>Coding part of MainActivity.java

```
package com.example.databaseapp;

import android.app.AlertDialog;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;


public class MainActivity extends AppCompatActivity {

EditText Rollno,Name,Marks;
 SQLiteDatabase db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Rollno=findViewById(R.id.Rollno);
        Name=findViewById(R.id.Name);
        Marks=findViewById(R.id.Marks);

        db=openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS student(rollno VARCHAR,name VARCHAR,marks
INT);");

    }
      public void onClickInsert(View view) {

          if (Rollno.getText().toString().trim().length() == 0 ||
```

```
            Name.getText().toString().trim().length() == 0 ||
            Marks.getText().toString().trim().length() == 0) {
          showMessage("Error", "Please enter all values");
          return;
        }
        db.execSQL("INSERT INTO student VALUES('" + Rollno.getText() + "','" + Name.getText() +
            "','" + Marks.getText() + "');");

        showMessage("Success", "Record added");
        clearText();

      }

      public void onClickDelete(View view) {

        if(Rollno.getText().toString().trim().length()==0)
        {
          showMessage("Error", "Please enter Rollno");
          return;
        }
        Cursor c=db.rawQuery("SELECT * FROM student WHERE rollno='"+Rollno.getText()+"'",
null);
        if(c.moveToFirst())
        {
          db.execSQL("DELETE FROM student WHERE rollno='"+Rollno.getText()+"'");
          showMessage("Success", "Record Deleted");
        }
        else
        {
          showMessage("Error", "Invalid Rollno");
        }
        clearText();
      }

      public void onClickUpdate(View view){

        if(Rollno.getText().toString().trim().length()==0)
        {
          showMessage("Error", "Please enter Rollno");
          return;
        }
        Cursor c=db.rawQuery("SELECT * FROM student WHERE rollno='"+Rollno.getText()+"'",
null);
        if(c.moveToFirst()) {
          db.execSQL("UPDATE student SET name='" + Name.getText() + "',marks='" +
Marks.getText() +
              "' WHERE rollno='"+Rollno.getText()+"'");
          showMessage("Success", "Record Modified");
        }
        else {
          showMessage("Error", "Invalid Rollno");
        }
        clearText();
      }
```

```
    public void onClickView(View view) {

        if(Rollno.getText().toString().trim().length()==0)
        {
            showMessage("Error", "Please enter Rollno");
            return;
        }
        Cursor c=db.rawQuery("SELECT * FROM student WHERE rollno='"+Rollno.getText()+"'",
null);
        if(c.moveToFirst())
        {
            Name.setText(c.getString(1));
            Marks.setText(c.getString(2));
        }
        else
        {
            showMessage("Error", "Invalid Rollno");
            clearText();
        }
    }

    public void onClickViewAll(View view) {
        Cursor c=db.rawQuery("SELECT * FROM student", null);
        if(c.getCount()==0)
        {
            showMessage("Error", "No records found");
            return;
        }
        StringBuffer buffer=new StringBuffer();
        while(c.moveToNext())
        {
            buffer.append("Rollno: "+c.getString(0)+"\n");
            buffer.append("Name: "+c.getString(1)+"\n");
            buffer.append("Marks: "+c.getString(2)+"\n\n");
        }
        showMessage("Student Details", buffer.toString());
    }
  public void showMessage(String title,String message)
  {
    AlertDialog.Builder builder=new AlertDialog.Builder(this);
    builder.setCancelable(true);
    builder.setTitle(title);
    builder.setMessage(message);
    builder.show();
  }
  public void clearText()
  {
    Rollno.setText("");
    Name.setText("");
    Marks.setText("");
    Rollno.requestFocus();
  }
}
```

=>Output of the program



10. Create an application to send SMS

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml, MainActivity.java and AndroidManifest.xml

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">


  <EditText
    android:id="@+id/etPhoneNumber"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="text"
```

```xml
        android:hint="Enter Phone Number"
       />

    <EditText
        android:id="@+id/etMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"
        android:inputType="text"
       />

    <Button
        android:id="@+id/btnSendSMS"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:onClick="onClick"
        android:text="Send SMS" />
</LinearLayout>
```

=>Coding part of MainActivity.java

```java
package com.example.sendsms;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
import android.Manifest;

public class MainActivity extends AppCompatActivity {


    private static final int SMS_PERMISSION_CODE = 101;
    EditText etPhoneNumber, etMessage;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etPhoneNumber = findViewById(R.id.etPhoneNumber);
        etMessage = findViewById(R.id.etMessage);

        if (!checkSMSPermission()) {
            requestSMSPermission();
        }
    }
    private boolean checkSMSPermission() {
         return ContextCompat.checkSelfPermission(this,
                Manifest.permission.SEND_SMS) == PackageManager.PERMISSION_GRANTED;
    }

    private void requestSMSPermission() {
         ActivityCompat.requestPermissions(this, new
                String[]{Manifest.permission.SEND_SMS}, SMS_PERMISSION_CODE);
    }
    public void onClick(View view) {
```

```java
        String strPhoneNumber = etPhoneNumber.getText().toString();
        String strMessage = etMessage.getText().toString();
        try {
            SmsManager smsManager = SmsManager.getDefault();
            smsManager.sendTextMessage(strPhoneNumber,null,strMessage,null, null);
            Toast.makeText(getApplicationContext(), "Message Sent", Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            Toast.makeText(getApplicationContext(), e.toString(), Toast.LENGTH_SHORT).show();
        }
    }
}
```

=>Coding part of AndroidManifest.xml: add uses-permission and uses-feature tags

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.SEND_SMS"/>
    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SendSMS">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

=>Output of the program

11. Create an application to send Email

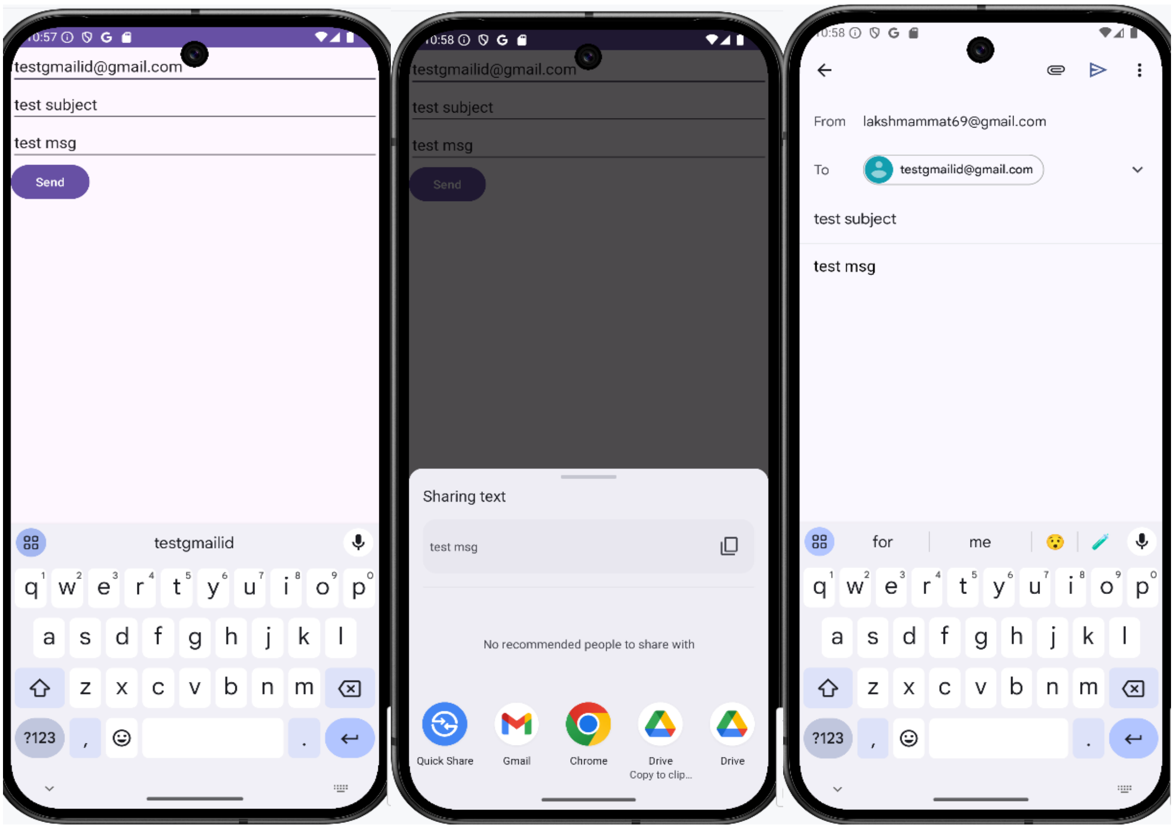
Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">
  <EditText
    android:id="@+id/etTo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="To"/>
  <EditText
    android:id="@+id/etSubject"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/etTo"
    android:hint="Subject"/>
  <EditText
    android:id="@+id/etMessage"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/etSubject"
    android:hint="Message"/>
  <Button
    android:id="@+id/btnSend"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="onClick"
    android:layout_below="@id/etMessage"
    android:text="Send"/>
</LinearLayout>
```

=>Coding part of MainActivity.java

```java
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity {
  EditText etTo, etSubject, etMessage;

  @Override
  protected void onCreate(Bundle savedInstanceState)
  {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    etTo = findViewById(R.id.etTo);
    etSubject = findViewById(R.id.etSubject);
    etMessage = findViewById(R.id.etMessage);
  }

  public void onClick(View v){
    String strTo = etTo.getText().toString().trim();
    String strSubject = etSubject.getText().toString().trim();
    String strMessage = etMessage.getText().toString().trim();
    Intent intent = new Intent(Intent.ACTION_SEND);
    intent.setType("text/plain");
    intent.putExtra(Intent.EXTRA_EMAIL, new String[]{strTo});
    intent.putExtra(Intent.EXTRA_SUBJECT, strSubject);
    intent.putExtra(Intent.EXTRA_TEXT, strMessage);
    if (intent.resolveActivity(getPackageManager()) != null)
    {
      startActivity(Intent.createChooser(intent, "Choose an email client"));
    }
  }
}
```



12. Create an Application to Display Map based on the Current/given location

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24("Nougat",Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">
  <fragment
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_alignParentTop="true"
    android:layout_alignParentBottom="true"
    android:layout_alignParentStart="true"
    android:layout_alignParentEnd="true" />
</RelativeLayout>
```

=>Coding part of MainActivity.java

```
package com.example.myapplication;
import android.os.Bundle;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
public class MainActivity extends AppCompatActivity implements OnMapReadyCallback {
  private GoogleMap mMap;
  private double latitude = 0.0;
  private double longitude = 0.0;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    // Obtain the SupportMapFragment and get notified when the map is ready to be used.
    SupportMapFragment mapFragment = (SupportMapFragment)
        getSupportFragmentManager()
            .findFragmentById(R.id.map);
    if (mapFragment != null) {
      mapFragment.getMapAsync(this);
    } else {
      Toast.makeText(this, "Map Fragment Not Found", Toast.LENGTH_SHORT).show();
    }
  }
  @Override
```

```
    public void onMapReady(@NonNull GoogleMap googleMap) {
        mMap = googleMap;
        // Add a marker at current or given location and move the camera
        LatLng location = new LatLng(latitude, longitude);
        mMap.addMarker(new MarkerOptions().position(location).title("Marker"));
        mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(location, 15));
    }
}
```

Coding part of AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <usespermission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <usespermission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <usespermission android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication"
        tools:targetApi="31">
        <metadata
            android:name="com.google.android.geo.API_KEY"
            android:value="YOUR_API_KEY_HERE" />

     <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

13. Create an Application with Login module. Check User name and password. On successful login change textview “Login Successful”. On Login fail alert using Toast “Login Fail”

Steps:
1. Click New Project, the New Project Dialog box appears.
2. Choose Empty Views Activity then click Next.
3. Specify the Name of your project, Select the Language as Java, and Select the SDK as API 24(“Nougat”,Android 7.0).Click Finish Button.
4. Update the following code in activity_main.xml and MainActivity.java

=>Coding part of Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

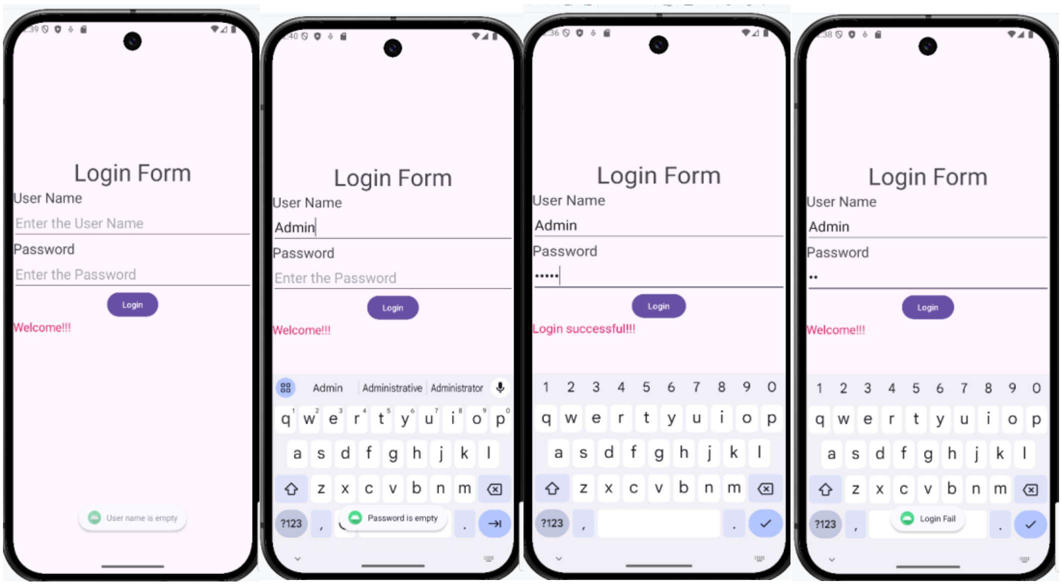
```
  android:orientation="vertical"
  tools:context=".MainActivity">

  <TextView
    android:id="@+id/tvTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:layout_marginTop="200dp"
    android:text="Login Form"
    android:textSize="40sp" />

  <TextView
    android:id="@+id/tvUserName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="User Name"
    android:textSize="24sp" />

  <EditText
    android:id="@+id/etUserName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the User Name"
    android:inputType="text"
    android:textSize="24sp" />

  <TextView
    android:id="@+id/tvPassword"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Password"
    android:textSize="24sp" />

  <EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the Password"
    android:inputType="textPassword"
    android:textSize="24sp" />

  <Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:onClick="onClick"
    android:text="Login" />

  <TextView
    android:id="@+id/tvMessage"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Welcome!!!"
    android:textColor="#E91E63"
    android:textSize="20sp" />

</LinearLayout>
```

=>Coding part of MainActivity.java

```
package com.example.loginapplication;

import android.os.Bundle;
```

```java
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import android.view.View;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;


public class MainActivity extends AppCompatActivity {

   EditText etUserName,etPassword;
   TextView tvMessage;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);

      etUserName=(EditText)findViewById(R.id.etUserName);
      etPassword=(EditText)findViewById(R.id.etPassword);
      tvMessage=(TextView)findViewById(R.id.tvMessage);
   }



public void onClick(View v)
{

   tvMessage.setText("Welcome!!!");
   if(etUserName.getText().toString().isEmpty())
   {
      Toast.makeText(this,"User name is empty", Toast.LENGTH_SHORT).show();
      return;
   }
   if(etPassword.getText().toString().isEmpty())
   {
      Toast.makeText(this,"Password is empty", Toast.LENGTH_SHORT).show();
      return;
   }
   if(etUserName.getText().toString().equals("Admin") &&
etPassword.getText().toString().equals("Admin"))
   {
      tvMessage.setText("Login successful!!!");
   }
   else
   {
      Toast.makeText(this, "Login Fail", Toast.LENGTH_SHORT).show();
   }
 }
}
```

=>OUTPUT OF THE PROGRAM



14. Learn to deploy Android applications

Steps to Deploy an Android Application

1. **Prepare App (use Program 1 Hello world for this program)**

2. **Generate Signed APK (Android Package Kit):**

   a. In Android Studio, navigate to Build > Generate Signed Bundle/APK.

   b. Follow the prompts to create a new keystore or use an existing one. A keystore is a binary file that contains a set of private keys.

   c. Configure the build type (release) and signing configuration.

   d. Generate the signed APK file.

3. **Test your signed APK**:

   a. Before distributing your app, test the signed APK to ensure that the signing process didn't introduce any issues.

   b. Install the APK on various devices and perform thorough testing.

   c. Release on Google Play Console:

   d. Sign in to the Google Play Console (https://play.google.com/apps/publish).

   e. Create a new app entry if this is your first release or select an existing app.

   f. Complete all the required information for the app listing, including the title, description, screenshots, and categorization

   g. Upload your signed APK file.

   h. Set pricing and distribution options.

i. Optimize your store listing for search and conversion.
Once everything is set, click the "Publish" button to release your app to the Google Play Store.

4. Other Distribution Channels (Optional):

- Besides Google Play, you can distribute your app through other channels such as Amazon Appstore, Samsung Galaxy Store, or third party app marketplaces.
- Each distribution channel may have its own requirements and submission process, so be sure to follow their guidelines.

5. Monitor and Update:

- Keep an eye on user feedback and app performance metrics through the Google Play Console.
- Regularly update your app to fix bugs, add new features, and improve user experience based on feedback.